

Integer Natural Evolution Strategies

Jacob de Nobel¹, Diederick Vermetten², Hao Wang¹,
Ofer M. Shir³, Michael Emmerich⁴, Thomas Bäck¹

¹ Leiden University, Leiden, The Netherlands

² Sorbonne Université, CNRS, LIP6, Paris, France

³ Tel-Hai - University of Kiryat Shmona in the Galilee, Israel

⁴ Faculty of Information Technology, University of Jyväskylä, Finland

Abstract. While contemporary Evolution Strategies handle integer optimization problems effectively, their adaptation mechanism is grounded in ℓ_2 -based Gaussian models, which are not native to the integer lattice. In contrast, the ℓ_1 -norm provides the natural measure of displacement on $\mathbb{Z}^n$, with the double geometric distribution as its canonical mutation operator. In this work, we derive a fully ℓ_1 -native step-size adaptation mechanism from first principles and propose an Integer Natural Evolution Strategy. We show that the DG distribution belongs to the exponential family, and that its sufficient statistic $|z|$ yields a natural-gradient signal for dispersion adaptation. By accumulating this signal via an evolution path, we obtain a fading-memory online estimator of the natural gradient, following Ollivier (2018). This establishes that DG-based step-size adaptation arises directly from the statistical structure of the mutation distribution, rather than as a discrete analog of continuous ES mechanisms. Empirical results on integer quadratic benchmarks show that INES learns meaningful coordinate-wise step-sizes and is competitive with integer-handling CMA-ES baselines. Its advantages are most visible in high-dimensional Ellipsoidal problems and in robust convergence at larger dimensions.

Keywords: Integer optimization, natural gradient, maximum-likelihood, evolution strategies.

1 Introduction

Integer Optimization focuses on formulating and solving models whose objective and constraint functions involve variables restricted to the integer lattice $\mathbb{Z}^n$. Research in this area is closely allied with Mixed-Integer (MI) Optimization (for a recent review see [20]), primarily because the vast majority of real-world integer problems arise in MI forms, requiring a blend of real- and discrete-valued variables. The discrete domain induces a different geometry and fundamentally alters the search-space landscape; specifically, the ℓ_1 norm captures the inherent grid-like connectivity more faithfully than the ℓ_2 norm, and discretization can transform unimodal continuous objectives into multimodal landscapes riddled with local traps [19]. Within the landscape of heuristic optimization, Evolution Strategies (ESs) [4] represent a highly effective class of randomized search

heuristics. They are distinguished from other Evolutionary Algorithms by their rigorous grounding in both biological principles and theoretical analysis and by their sophisticated self-adaptation of search parameters [4]. In addition, ESs are invariant under monotonic transformations of the objective function and angle-preserving transformations of the decision space, which includes rotations and translations [4]. While predominantly applied to continuous domains, the ES framework has been extended to treat integer and mixed spaces [18,7]. To date, however, these adaptations largely rely on truncated variants of mutation distributions originally conceived for continuous optimization [13]. In this work, we move beyond these truncated approximations to investigate the effectiveness of the *Double Geometric* (DG) distribution [17]. As the discrete counterpart to the continuous Laplace distribution [11], the DG has long been recognized as a natural choice for such domains. Recently, [19] reaffirmed its advantages over the commonly used *Truncated Normal* (TN) and demonstrated the feasibility of capturing correlations between integer variables.

In the continuous domain, however, research has progressed from classical self-adaptation to *derandomized* parameter control, which aims to update strategy parameters without any independent stochastic variation. This culminated in the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [9], which is the current state of the art for continuous black-box optimization. Despite its success, analogous derandomized mechanisms have not been systematically developed for (mixed) integer optimization. This gap stems from the difficulty of modeling discrete correlations and the lack of a natural analog of the covariance matrix on the integer lattice. Consequently, and somewhat controversially, one of the most competitive strategies for integer optimization remains the direct application of CMA-ES with only minimal modifications [13], despite its fundamental design for continuous manifolds.

Another reason to move from Euclidean to lattice-native step-size control is that a fixed ℓ_2 radius does not determine the effective mutation length after applying the componentwise rounding map $\text{rnd}[\cdot]$ to $\mathbb{Z}^n$. Indeed, points on the same Euclidean shell may induce vastly different discrete ℓ_1 lengths after rounding, depending on how their mass is distributed across coordinates. Thus, Euclidean truncation alone is not a reliable proxy for the mutation strength (See [5]).

Against this background, we address the following fundamental question:

Research Question. Can step-size adaptation in integer ES be derived from first principles using an ℓ_1 -native mutation model and information-geometric principles, and does this yield competitive performance on $\mathbb{Z}^n$?

Contributions. We provide a principled approach to step-size adaptation in Integer Evolution Strategies (IESs):

- **Natural-gradient derivation:** We show that the double geometric (DG) distribution forms an exponential family whose sufficient statistic $|z|$ yields a natural-gradient signal for step-size adaptation, providing a first-principles justification for ℓ_1 -native updates.
- **Integer Natural Evolution Strategy:** We develop a $(1, \lambda)$ integer natural evolution strategy that combines DG-based mutation with a fading-memory

natural-gradient estimator, resulting in a simple and stable ℓ_1 -native adaptation mechanism.

- **Empirical evaluation:** We demonstrate on integer quadratic benchmarks that the proposed method learns meaningful coordinate-wise step-sizes and delivers competitive, robust performance against both full- and separable-integer CMA-ES baselines.

Paper Organization. The remainder of this paper is organized as follows. Section 2 introduces the DG distribution and IES fundamentals. Section 3 presents the ℓ_1 -native natural gradient and the resulting IES. Section 4 provides the empirical evaluation, and Section 5 concludes.

2 Preliminaries

Evolution Strategies (ESs) are the predominant heuristics for continuous optimization, with derandomized variants, most notably CMA-ES [9], representing the state-of-the-art. In ES, a current decision vector $\mathbf{x}^{(t)}$ is perturbed by a mutation vector $\mathbf{z}$ to generate a candidate solution

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} + \mathbf{z},$$

where $\mathbf{z}$ is drawn from a parameterized search distribution. Depending on the model, the components z_i may be treated as independent or correlated. A defining feature of ES is the *adaptation of the mutation distribution itself*. The parameters of the search distribution, such as the step size σ for the multivariate normal distribution, are updated over time based on the observed fitness values, allowing the algorithm to adapt its search behavior to the underlying landscape.

Natural Evolution Strategies (NESs) [21] take a principled distributional view of this adaptation: rather than adapting strategy parameters heuristically, NESs explicitly maintain a parameterized search distribution $\pi(\mathbf{z} \mid \theta)$ and minimize the expected cost

$$J(\theta) := \mathbb{E}_\theta[f(\mathbf{x} + \mathbf{z})]$$

by following the *search gradient*, which reads

$$\nabla_\theta J(\theta) = \mathbb{E}_\theta[f(\mathbf{x} + \mathbf{z}) \nabla_\theta \log \pi(\mathbf{z} \mid \theta)], \quad (1)$$

and can be statistically approximated via a population of mutations $\{\mathbf{z}_k\}_{k=1}^\lambda$ as

$$\nabla_\theta J(\theta) \approx \frac{1}{\lambda} \sum_{k=1}^\lambda f(\mathbf{x} + \mathbf{z}_k) \nabla_\theta \log \pi(\mathbf{z}_k \mid \theta).$$

Grounded in Information Geometry [2], NESs precondition this gradient by the inverse Fisher information matrix $\mathcal{F}(\theta)^{-1}$, yielding the *natural gradient*

$$\tilde{\nabla}_\theta J(\theta) := \mathcal{F}(\theta)^{-1} \nabla_\theta J(\theta),$$

which is invariant under reparameterizations of π . The resulting update rule,

$$\theta \leftarrow \theta - \eta \mathcal{F}(\theta)^{-1} \nabla_{\theta} J(\theta), \quad (2)$$

with η serving as a learning rate, steers the search distribution toward regions of lower objective value in a geometrically principled manner [21].

2.1 Going Discrete: The Double Geometric Distribution

Extending ESs to Integer Optimization requires rethinking both the mutation distribution and step-size control. While Gaussian distributions and ℓ_2 -based measures are natural in continuous spaces, they do not align with the geometry of the integer lattice $\mathbb{Z}^n$. Instead, ℓ_1 -based quantities and lattice-native mutation operators provide a more appropriate foundation [19]. This choice is supported by information-theoretic considerations: Rudolph [17] showed that, under a constraint on the expected absolute step-size, the DG distribution is the maximum-entropy distribution on $\mathbb{Z}$. The magnitude of mutations on the integer lattice is naturally quantified by the expected ℓ_1 -norm

$$S := \mathbb{E}[\|\mathbf{z}\|_1] = \sum_{i=1}^n \mathbb{E}[|Z_i|],$$

which characterizes the overall mutation magnitude in $\mathbb{Z}^n$. In the following, we primarily work with the per-coordinate quantities $\delta_i := \mathbb{E}[|Z_i|]$, which serve as the fundamental step-size parameters.

The DG distribution can be constructed by drawing two independent geometric random variables $G_1, G_2 \sim \text{Geom}(p)$ with

$$\Pr\{G_j = k\} = p(1-p)^k, \quad k \in \mathbb{N}_0,$$

and setting $Z = G_1 - G_2$. Its probability mass function (PMF) is

$$\Pr\{Z = k\} = \frac{p}{2-p} (1-p)^{|k|}, \quad k \in \mathbb{Z}. \quad (3)$$

It is convenient to *reparameterize* using $q := 1-p$, $q \in (0, 1)$ – yielding the form $\Pr\{Z = k\} \propto q^{|k|}$, which we adopt throughout the paper.

The DG distribution has the following key statistics:

$$\mathbb{E}[Z] = 0, \quad \text{Var}[Z] = \frac{2q}{(1-q)^2}. \quad (4)$$

The expected absolute step-size of a single DG variable is related to q via the following invertible *map*

$$\delta(q) = \mathbb{E}[|Z|] = \frac{2q}{1-q^2} \iff q(\delta) = \frac{\delta}{\sqrt{1+\delta^2}+1}. \quad (5)$$

In the isotropic case, where all coordinates share the same parameter, the global step-size for the DG distribution satisfies $S_{\text{DG}} = n \cdot \delta$.

The ℓ_1 -based notion of step-size is not merely a modeling preference, but reflects a genuine geometric discrepancy between Euclidean and lattice-native mutation length. To see this, let $\text{rnd}[\mathbf{x}]$ denote componentwise rounding of $\mathbf{x} \in \mathbb{R}^n$ to the nearest integer vector, with exact halves rounded downward, and consider the rounded ℓ_1 length

$$L(\mathbf{x}) := |\text{rnd}[\mathbf{x}]|_1 \quad \text{subject to} \quad |\mathbf{x}|_2 = d.$$

Then $L(\mathbf{x})$ is not determined by the Euclidean radius d alone. If the Euclidean mass is spread sufficiently thinly across coordinates, for example uniformly over more than $4d^2$ coordinates, then each component has magnitude below $1/2$, so that $\text{rnd}[\mathbf{x}] = \mathbf{0}$ and hence $L(\mathbf{x}) = 0$. By contrast, for the balanced direction $\mathbf{x} = (d/\sqrt{n}, \dots, d/\sqrt{n})$, one obtains

$$L(\mathbf{x}) = n \left[\frac{d}{\sqrt{n}} - \frac{1}{2} \right].$$

In particular, this quantity is at least n whenever $n < 4d^2$, and at least $2n$ whenever $n \leq d^2/4$. Thus, vectors lying on the same ℓ_2 shell can correspond, after rounding, to a large range of different effective ℓ_1 mutation lengths. This is one reason why the expected absolute step-size $S = \mathbb{E}[|\mathbf{z}|_1]$ is the natural quantity to control on $\mathbb{Z}^n$, whereas Euclidean truncation does not by itself characterize the mutation strength seen by the lattice. A detailed derivation is provided as Supp. Material [5].

2.2 Perspective: Exponential Families and Information Geometry

We now situate the DG distribution within the framework of exponential families and Information Geometry. This perspective provides a direct link between distribution parameters and the step-size quantities $\delta_i = \mathbb{E}[|Z_i|]$ introduced earlier. Modeling discrete search updates through this lens has strong precedent in Estimation of Distribution Algorithms (EDAs), where Information Geometry provides a principled basis for probability model updates [12]. Under this geometric view, parameter updates correspond to steepest ascent in distribution space with respect to the Fisher–Rao metric, i.e., natural gradient ascent [2]. This mirrors the continuous setting, where covariance matrix adaptation in Gaussian ESs can be interpreted as natural gradient ascent on the Gaussian (ℓ_2) manifold [1]. Our framework transposes this principle to the discrete (ℓ_1) domain.

A PMF $\pi(\mathbf{z} \mid \boldsymbol{\theta})$ belongs to the exponential family if it can be written as

$$\Pr(\mathbf{z} \mid \boldsymbol{\theta}) = h(\mathbf{z}) \exp(\boldsymbol{\theta}^\top \boldsymbol{\phi}(\mathbf{z}) - A(\boldsymbol{\theta})), \quad (6)$$

where $\boldsymbol{\theta}$ denotes the natural parameters, $\boldsymbol{\phi}(\mathbf{z})$ the sufficient statistic, $h(\mathbf{z})$ the base measure, and $A(\boldsymbol{\theta})$ the log-partition function.

The DG distribution (3) admits this form coordinate-wise as

$$\Pr(z_i \mid q_i) = \frac{1 - q_i}{1 + q_i} \exp(|z_i| \ln q_i), \quad (7)$$

which identifies the natural parameter $\theta_i = \ln q_i$, sufficient statistic $\phi(z_i) = |z_i|$, base measure $h(z_i) = 1$, and log-partition function

$$A(\theta_i) = \ln\left(\frac{1 + q_i}{1 - q_i}\right) = \ln\left(\frac{1 + e^{\theta_i}}{1 - e^{\theta_i}}\right).$$

Using the inversion map (5), the exponential-family structure yields the identities

$$A'(\theta_i) = \delta_i, \quad A''(\theta_i) = \text{Var}[|z_i|] = \delta_i \sqrt{1 + \delta_i^2}, \quad (8)$$

establishing that the natural parameters directly control the coordinate-wise step-sizes, while $A''(\theta_i)$ provides the Fisher information governing the scaling of natural gradient updates.

2.3 Score Function and Maximum Likelihood Estimation

Since $\phi(z) = |z|$ is the sufficient statistic of the DG family (Section 2.2), the absolute deviation provides the natural signal for step-size adaptation on the integer lattice. For a single observation z , the log-likelihood reads

$$\mathcal{L}(\theta; z) = \phi(z)\theta - A(\theta).$$

By differentiating with respect to θ , and by using the exponential-family identity $A'(\theta) = \delta = \mathbb{E}_\theta[\phi(Z)]$ [3], one obtains the *score function*

$$\boxed{\nabla_\theta \mathcal{L} = \phi(z) - A'(\theta) = |z| - \delta.} \quad (9)$$

Maximum likelihood estimation. For n i.i.d. observations $z^{(1)}, \dots, z^{(n)}$ drawn from a common $\text{DG}(p)$, the total log-likelihood is

$$\mathcal{L}(\theta; z^{(1)}, \dots, z^{(n)}) = \theta \sum_{j=1}^n \phi(z^{(j)}) - nA(\theta).$$

By the Fisher–Neyman factorization theorem, the sum $\sum_{j=1}^n \phi(z^{(j)})$ is the sufficient statistic for θ . The maximum-likelihood estimator $\hat{\theta}$ must satisfy the **moment matching** condition [14]:

$$A'(\hat{\theta}) = \frac{1}{n} \sum_{j=1}^n \phi(z^{(j)}) \iff \mathbb{E}_{\hat{\theta}}[\phi(Z)] = \bar{\phi},$$

where $\bar{\phi}$ is the empirical mean of the sufficient statistic.

Crucially, no information beyond $\bar{\phi}$ is required to estimate the dispersion of DG mutations. Neither the sample variance nor higher-order moments are necessary; **the first moment of the absolute deviations is a complete descriptor** for the ℓ_1 -native lattice dispersion.

3 Integer Natural Evolution Strategy by First Principles

In the $(1, \lambda)$ setting, the selected mutation $\mathbf{z}_s$ provides one observation per generation. For coordinate i , the corresponding DG score (9) is

$$\nabla_{\theta_i} \mathcal{L} = \phi(z_{s,i}) - \mathbb{E}_{\theta_i}[\phi(Z_i)] = |z_{s,i}| - A'(\theta_i). \quad (10)$$

This is the score (log-likelihood gradient) of the DG dispersion parameter at that coordinate: it is positive when the observed step exceeds the model expectation, indicating that the dispersion should increase, and negative when it falls short, indicating that it should decrease. This parallels the Gaussian/ ℓ_2 setting, where the sufficient statistic is z^2 , and cumulative step-size adaptation [9] is driven by $\|\mathbf{z}\|_2^2 - \mathbb{E}[\|\mathcal{N}(\mathbf{0}, \mathbf{I})\|_2^2]$.

Fisher information and the natural gradient form. The natural gradient $\tilde{\nabla}_{\theta_i}$ is obtained by scaling the score $\nabla_{\theta_i} \mathcal{L}$ by the inverse Fisher information $\mathcal{F}(\theta_i)^{-1}$. Since the DG distribution is a one-parameter exponential family coordinatewise, the Fisher information reduces to the variance of the sufficient statistic:

$$\mathcal{F}(\theta_i) = \text{Var}[\phi(Z_i)] = A''(\theta_i).$$

Thus, the natural gradient takes the explicit form:

$$\tilde{\nabla}_{\theta_i} = \frac{\phi(z_{s,i}) - \mathbb{E}[\phi(Z_i)]}{\text{Var}[\phi(Z_i)]} = \frac{|z_{s,i}| - \mathbb{E}[|Z_i|]}{\text{Var}[|Z_i|]}. \quad (11)$$

Statistical-estimation perspective. The innovation $|z_{s,i}| - \mathbb{E}[|Z_i|]$ is the score of the DG exponential family, and thus the canonical signal for estimating its dispersion parameter [3]. By the Fisher–Neyman factorization theorem [14], no information beyond the sample value of $|z_{s,i}|$ is required to update the model. Ollivier [15] showed that exponentially discounted accumulation of the natural gradient for exponential-family parameters yields an online natural-gradient estimator, equivalent to an extended Kalman filter. This motivates defining the evolution path ϕ as a *fading-memory accumulator of the natural gradient*:

$$\phi \leftarrow (1 - c) \phi + c \cdot \tilde{\nabla}_{\theta}, \quad (12)$$

where $\tilde{\nabla}_{\theta}$ denotes the coordinate-wise natural gradient defined in (11). The normalization by the Fisher information ensures that path entries across coordinates contribute on a common, Fisher-calibrated scale; a practical regularization is discussed in Remark 4.

Remark 1. We can interpret the fading-memory accumulator/evolution path in Eq. (12) through the lens of dynamical systems and show that it stabilizes the learning of the DG parameters. Let $g(\theta) = \mathcal{F}(\theta)^{-1} \nabla_{\theta} \mathcal{L}$ denote the natural gradient. Consider a (local) maximum θ^* of $\mathcal{L}$, i.e., $g(\theta^*) = 0$ and $J := \text{dg}(\theta^*) < 0$.

First, since the fading-memory accumulator ϕ estimates the θ^* in an online manner, we can understand ϕ_k as a momentum in iteration k :

$$\theta_{k+1} = \theta_k + \eta \phi_{k+1}, \quad \phi_{k+1} = (1 - c)\phi_k + cg(\theta_k), \quad k = 0, 1, 2, \dots$$

where $\eta > 0$ is the learning rate. Next, define $\Delta_k := \theta_k - \theta^*$ and linearize the natural gradient around θ^* , i.e., $g(\theta_k) = J\Delta_k$. We have the linear system $\phi_{k+1} = (1 - c)\phi_k + cJ\Delta_k$, $\Delta_{k+1} = \Delta_k + \eta\phi_{k+1}$. The stability of the system is determined by the spectrum of J . Let ρ be an eigenvalue of J with corresponding eigenvector v . Restricting the dynamics to this direction by writing $\Delta_k = x_kv$ and $\phi_k = y_kv$, we obtain the scalar recursion $y_{k+1} = (1 - c)y_k + c\rho x_k$ and $x_{k+1} = x_k + \eta y_{k+1}$ which can be written as

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = \begin{pmatrix} 1 + \eta c \rho \eta (1 - c) \\ c\rho & 1 - c \end{pmatrix} \begin{pmatrix} x_k \\ y_k \end{pmatrix}.$$

The system is stable along v if and only if the eigenvalues of this matrix lie inside the unit disk, which yields the condition

$$\eta < \frac{2(2 - c)}{c|\rho|}.$$

For comparison, the plain gradient system $\Delta_{k+1} = \Delta_k + \eta J\Delta_k$ is stable under the stricter condition $\eta < 2/|\rho|$. Since $2(2 - c)/c > 2$ for all $0 < c < 1$, the accumulator enlarges the admissible range of learning rates. In other words, it stabilizes the dynamics and permits more aggressive updates. This effect is analogous to momentum methods in continuous optimization, where temporal smoothing improves stability of the update dynamics.

Verification We formalize this interpretation using the result of Ollivier [15].

Proposition 1. *Let $\theta_i = \ln q_i$ be the DG natural parameter for coordinate i , and let $\mathcal{F}(\theta_i) = A''(\theta_i) = \text{Var}_{\theta_i}[\phi(Z_i)]$ denote the corresponding Fisher information. The evolution path update*

$$\phi_i \leftarrow (1 - c)\phi_i + c \cdot \frac{\phi(z_{s,i}) - \mathbb{E}_{\theta_i}[\phi(Z_i)]}{\mathcal{F}(\theta_i)}$$

is a fading-memory online natural gradient estimator for θ_i [15, Proposition 3], with learning rate c .

Proof. The DG distribution is a one-parameter exponential family with natural parameter θ_i , sufficient statistic $\phi(z) = |z|$, and log-partition function $A(\theta_i)$. For a single observation $z_{s,i}$, the score is $\partial \mathcal{L} / \partial \theta_i = \phi(z_{s,i}) - A'(\theta_i) = |z_{s,i}| - \mathbb{E}_{\theta_i}[|Z_i|]$.

The Fisher information for this family is the variance of the sufficient statistic, $\mathcal{F}(\theta_i) = A''(\theta_i) = \text{Var}_{\theta_i}[|Z_i|]$. The natural gradient step for θ_i is therefore $\mathcal{F}(\theta_i)^{-1}(|z_{s,i}| - \mathbb{E}_{\theta_i}[|Z_i|])$. Accumulating this signal with exponential discounting at rate c yields the stated update for ϕ_i , which coincides with the fading-memory online natural gradient of [15, Proposition 3] applied coordinate-wise to the DG family. $\square$

Consequently, DG-based cumulation follows directly from the exponential-family structure of the mutation model. The score provides the maximum-likelihood innovation, whose exponentially discounted accumulation constitutes a natural gradient estimator in the sense of [15]. This ensures that the evolution path ϕ is grounded in the Information Geometry of the underlying lattice distribution.

Score function for the location parameter. For completeness, we justify the center-of-mass update $\mathbf{m} \leftarrow \mathbf{m} + \mathbf{z}_s$ from a maximum-likelihood perspective. Given an observation $x_i = m_i + z_i$, where $z_i \sim \text{DG}(p_i)$, the log-likelihood as a function of the location parameter m_i is $\mathcal{L}(m_i) = \theta_i |x_i - m_i| + \text{const}$. The corresponding score (for $z_{s,i} \neq 0$) is

$$\frac{\partial \mathcal{L}}{\partial m_i} = -\text{sign}(z_{s,i}) \theta_i. \quad (13)$$

Since $\theta_i = \log q_i < 0$, this yields $|\theta_i| \text{sign}(z_{s,i})$, i.e., a direction aligned with the selected mutation. Thus, the update $\mathbf{m} \leftarrow \mathbf{m} + \mathbf{z}_s$ follows the maximum-likelihood gradient direction for the location parameter. The effective step length is governed by the dispersion parameters, which are adapted via the evolution path ϕ .

The DG exponential-family structure thus provides principled maximum-likelihood signals for both location and dispersion: the former through $\text{sign}(z_{s,i})$, and the latter through the residual $\phi(z_{s,i}) - \mathbb{E}[\phi(Z_i)]$.

A Concrete $(1, \lambda)$ -INES

We now combine the preceding derivations into a single algorithmic framework. The method maintains a center-of-mass $\mathbf{m} \in \mathbb{Z}^n$, a vector of expected absolute step-sizes $\delta \in \mathbb{R}_{>0}^n$, and a cumulative path $\phi \in \mathbb{R}^n$. Initially, $\delta_i = \delta_0$ for all i and $\phi = \mathbf{0}$. The following block summarizes a complete generation of a $(1, \lambda)$ -INES.

λ offspring are independently DG-sampled and evaluated:

- 1: **for** $k = 1, \dots, \lambda$ **do**
- 2: Sample $\mathbf{z}_k \sim \bigotimes_{i=1}^n \text{DG}(p_i(\delta_i))$
- 3: $\mathbf{x}_k \leftarrow \mathbf{m} + \mathbf{z}_k$, evaluate $f(\mathbf{x}_k)$
- 4: **end for**
- 5: $s \leftarrow \arg \min_k f(\mathbf{x}_k)$
- 6: $\mathbf{m} \leftarrow \mathbf{x}_s$

Fisher-normalized centered sufficient statistic as the adaptation signal:

- 1: $g_i \leftarrow \frac{|z_{s,i}| - \delta_i}{\max(\delta_i \sqrt{1 + \delta_i^2}, 1)}$
- 2: $\phi_i \leftarrow (1 - c)\phi_i + \sqrt{c(2 - c)} g_i$

Parameters follow the accumulated natural gradient:

- 1: $\delta_i \leftarrow \delta_i \exp(\eta \phi_i)$

Remark 2 (Direct step-size parametrization). A direct update in the natural parameter $\theta_i = \log q_i$ is problematic, as the domain constraint $q_i \in (0, 1)$ implies $\theta_i < 0$. Additive updates in θ do not preserve this constraint and may lead to invalid parameters ($q_i \geq 1$), at which point the DG moments diverge. To avoid explicit clipping, we instead parameterize the mutation strength directly via the expected absolute step size δ_i and perform updates in this space. Since $\delta_i > 0$ by definition, the update rule preserves positivity and thus always induces a valid DG parameter.

Remark 3 (Path normalization). The factor $\sqrt{c(2-c)}$ keeps the variance of the cumulative path ϕ independent of the time constant c . Without this scaling, longer memory (smaller c) would shrink ϕ , effectively reducing the learning rate. The normalization, therefore, decouples the adaptation time scale from the update magnitude.

Remark 4 (Practical normalization). The normalization of g_i corresponds to scaling by the Fisher information of the DG family, yielding a natural gradient update [15, Section 2.2]. This removes the dependence of the update magnitude on the local curvature of the model. In practice, when the distribution becomes highly concentrated, the variance, i.e. $\delta_i \sqrt{1 + \delta_i^2}$, becomes small, leading to excessively large updates. We therefore apply a lower bound via $\max(\cdot, 1)$, which acts as a Tikhonov-style regularization and prevents numerical instability.

ℓ_1 -Nativeness and Geometric Separation It is instructive to distinguish between the two geometric layers underlying the proposed algorithm.

Data-space geometry. Mutation magnitudes on the integer lattice $\mathbb{Z}^n$ are measured entirely in terms of absolute deviations. The resulting adaptation signal is based on the centered sufficient statistic $|z_i| - \mathbb{E}[|Z_i|]$, making the method fully ℓ_1 -native. The first moment $\delta_i = \mathbb{E}[|Z_i|]$ therefore acts as the natural scale parameter, and neither squared steps nor Euclidean norms enter the search dynamics.

Statistical-manifold geometry. While the search itself is ℓ_1 -based, the adaptation signal is preconditioned by the Fisher information of the DG family. This introduces a second moment that reflects the local curvature of the statistical model and yields a natural gradient update. Crucially, this second moment acts only on the scalar quantity $|Z_i|$ and calibrates parameter updates; it does not affect how distances are measured in $\mathbb{Z}^n$.

4 Numerical Observations

To empirically validate our proposed approach, we conduct a series of numerical procedures to calibrate and test it⁵. We consider axis-aligned convex benchmark

⁵ Due to space limitations, the complete experiments are in our Supp. Material [5]

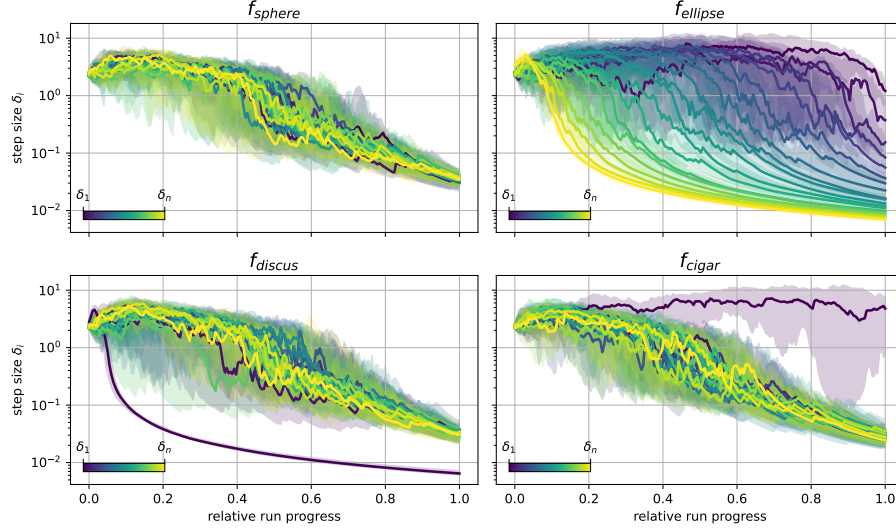


Fig. 1: Evolution of coordinate-wise step-sizes δ_i over relative run progress for $n = 20$. Solid lines show the median over 25 runs, with shaded areas indicating the interquartile range. Colors correspond to coordinate indices.

models induced by weighted quadratic forms on the integer domain. For each independent instance, the optimum $\mathbf{x}^*$ is sampled uniformly from $[-50, 50]^n$, and all algorithms are initialized from the same starting point, sampled from the same domain. The objective is

$$f(\mathbf{x}) = \|\mathbf{W}(\mathbf{x} - \mathbf{x}^*)\|_2, \quad (14)$$

where $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ controls the conditioning. We study four benchmark classes:

$$\begin{aligned} \text{Sphere} : w_i &= 1 \quad \forall i, \\ \text{Ellipse} : w_i &= 10^{4(i-1)/(n-1)}, \\ \text{Discus} : w_1 &= 10^4, w_i = 1 \text{ for } i > 1, \\ \text{Cigar} : w_1 &= 1, w_i = 10^4 \text{ for } i > 1. \end{aligned}$$

Parameter Calibration We calibrated our proposed INES using a *grid search* over the Sphere and Ellipsoid functions across multiple problem dimensionalities. Based on this, we fit a simple scaling law to the optimal tradeoff points, resulting in parameter settings $c(n) = 1 - \frac{1.5}{n}$ and $\eta(n) = \left(\frac{2}{n}\right)^{1/3}$, used throughout our experiments.

4.1 Analysis of Learned Step Sizes

We first focus on how the proposed method adapts coordinate-wise step-sizes. For each function and dimension $n \in \{2, 3, 5, 10, 20, 40, 100\}$, we perform 25

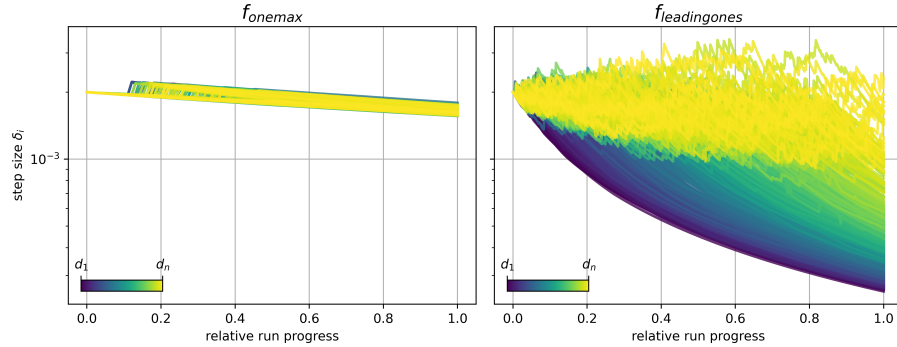


Fig. 2: Evolution of coordinate-wise step-sizes δ_i on ONEMAX and LEADINGONES for $n = 500$. Lines show the median over 25 runs; colors indicate coordinate index. The initial step-size is $\delta_i = 1/n$.

independent runs, each terminated upon reaching the optimum. We analyze the behavior of the raw step-sizes $\delta_i(t)$ directly. To compare runs of different lengths, trajectories are resampled to a common relative time scale $\tau \in [0, 1]$.

Dynamics of step-sizes. Figure 1 shows the evolution of step-sizes for $n = 20$. On the Sphere function, all coordinates evolve similarly over time, reflecting the absence of directional structure. On the Ellipse, a clear separation emerges, with step-sizes adapting to the underlying conditioning. On the Discus and Cigar functions, the method successfully identifies the distinguished direction: the corresponding coordinate exhibits a markedly different scale (suppressed for Discus, amplified for Cigar), while the remaining coordinates evolve in a largely symmetric manner. Despite the stochasticity of the updates, the qualitative behavior is consistent across runs and dimensionalities (see [5]) and altogether validates our INES design.

As an additional qualitative check, we inspect the learned step-sizes on the pseudo-Boolean benchmarks ONEMAX and LEADINGONES. To apply DG mutations on the binary domain, integer mutations are cyclically mapped to $\{0, 1\}^n$ by $(\mathbf{x} + \mathbf{z}) \bmod 2$. These experiments are not intended as a full PBO benchmark study, but illustrate (see Figure 2) how the same adaptation mechanism behaves on problems with different coordinate structures. On ONEMAX, the learned step-sizes remain comparatively uniform, whereas on LEADINGONES a pronounced coordinate-dependent pattern emerges.

4.2 Performance on Integer Quadratic Models

After confirming that the step size matches expected behavior, we move on to benchmark optimization performance. The experimental protocol follows the same setup as the previous experiment. As reference methods, we consider two

Table 1: Expected running time (ERT) on the four benchmark functions over 25 independent runs. Lower is better; ∞ indicates that the target was not reached within the budget of $10^4 \cdot n$ evaluations in any run. Best values per function–dimension pair are shown in boldface.

Function	Method	2	3	5	10	20	40	100
Sphere	INES	231	329	511	1,238	2,865	7,358	20 904
	CMA-IH	115	217	423	933	2,200	4 342	∞
	CMA-IH-SEP	108	181	345	744	1 736	4 343	88 395
Ellipse	INES	266	437	626	1 288	3 117	8 402	29 488
	CMA-IH	152	588	985	3 684	13 965	49 498	∞
	CMA-IH-SEP	171	344	806	2 070	5 246	16 024	∞
Discus	INES	287	461	668	1 389	3 140	7 927	23 244
	CMA-IH	173	297	647	1 547	3 712	8 232	∞
	CMA-IH-SEP	172	292	548	1 118	2 496	4 635	24 075 579
Cigar	INES	260	331	518	1 111	2 821	7 184	22 717
	CMA-IH	156	277	679	1 737	5 013	21 351	667 002
	CMA-IH-SEP	164	231	449	919	2 076	4 193	13 116

integer-handling variants of CMA-ES. The first is CMA-IH [13], which applies TN-based mutations over the integer lattice while adapting the CMA-ES mechanism to this discrete domain. The second is its separable variant, denoted CMA-IH-SEP, which restricts the covariance matrix to its diagonal elements [16]. This separable baseline is particularly relevant here because the current INES variant also adapts only coordinate-wise step-sizes. We use the default population size for each dimensionality as recommended in the CMA-ES literature, without additional parameter tuning, and set the initial step size to $\sigma_0 = 100/n$. This choice provides a common initialization across dimensions, but the sensitivity of the CMA-based baselines to σ_0 is not studied here and remains a limitation of the empirical comparison. To obtain an initial variance of σ_0^2 for the DG distribution, we set $\delta_0 = \sigma_0^2 / \sqrt{2\sigma_0^2 + 1}$ for INES.

Results Table 1 reports the expected running time (ERT) across dimensionalities [8]. All finite ERT values are based on 25 successful runs, except for CMA-IH-SEP on Discus at $n = 100$ (one success) and CMA-IH on Cigar at $n = 100$ (22 successes); these ERTs may exceed the per-run budget because failed runs contribute their full budget. On the Sphere function, the CMA-based baselines are faster in low and moderate dimensions, but at $n = 100$ CMA-IH fails to reach the target, and CMA-IH-SEP requires substantially more evaluations than INES. INES remains robust in high dimensions, but the lack of a global step-size component hinders performance on the isotropic Sphere model. INES shows its clearest advantage on the Ellipse function, where it obtains the best ERT from $n = 5$ onward and is the only method that reaches the target at $n = 100$. In fact,

even for a very large $n = 1000$ (not included in the Table), INES consistently finds the global optimum for the Ellipse model with an ERT of 399 483. This supports the step-size analysis above, indicating that the coordinate-wise DG adaptation captures the dominant conditioning structure. On Discus and Cigar, CMA-IH-SEP is faster on most lower- and moderate-dimensional settings, especially on Cigar, but also deteriorates strongly on Discus at $n = 100$. Overall, these results show that INES is an effective lattice-native solver with robust high-dimensional behavior.

5 Conclusion

We introduced the $(1, \lambda)$ Integer Natural Evolution Strategy (INES), a fully ℓ_1 -native approach to step-size adaptation on the integer lattice. The method combines double geometric (DG) mutations with a natural-gradient update derived from the exponential-family structure of the DG distribution. In this setting, the sufficient statistic $\phi(z) = |z|$ yields the adaptation signal $|z_i| - \mathbb{E}[|Z_i|]$, whose Fisher-scaled accumulation provides an online estimator of the natural gradient. For a practical algorithm, we use a fading-memory evolution path and directly parametrize the expected absolute step size $\delta_i = \mathbb{E}[|Z_i|]$, thereby avoiding validity issues arising from unconstrained updates of the natural parameter θ . This yields a simple, numerically stable adaptation mechanism consistent with the geometry of $\mathbb{Z}^n$.

Empirically, INES performs competitively on integer quadratic benchmarks and learns coordinate-wise step-sizes that reflect the underlying conditioning. Additional qualitative experiments on pseudo-Boolean benchmarks suggest that the same mechanism induces problem-dependent step-size dynamics on binary domains. Its strongest results appear for the Ellipse function and in several high-dimensional settings where the CMA-based baselines fail or perform poorly. At the same time, the current formulation is limited to coordinate-wise adaptation and lacks both a global step-size component and dependencies between variables, which likely explains its weaker performance on simple isotropic landscapes.

Overall, DG-based step-size adaptation is not an ad hoc discrete analog of continuous ES mechanisms, but follows directly from the statistical structure of the underlying mutation distribution. Future work includes convergence-rate analysis and extensions to multi-parent recombination. On the algorithmic side, the natural next steps are to integrate a global step-size component and to model dependencies between variables. Empirically, broader evaluations should include rotated, non-separable, and multimodal benchmarks, as well as comparisons with recent mixed-integer optimizers such as CMA-ES with Margin [6] and DX-NES-ICI [10].

Acknowledgments. Diederick Vermetten acknowledges funding by the European Union (ERC, “dynaBBO”, grant no. 101125586).

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Akimoto, Y., Nagata, Y., Ono, I., Kobayashi, S.: Bidirectional relation between CMA evolution strategies and natural evolution strategies. In: PPSN XI. pp. 154–163. Springer Berlin Heidelberg (2010)
2. Amari, S.I.: Natural gradient works efficiently in learning. *Neural Comput.* **10**(2), 251–276 (1998)
3. Barndorff-Nielsen, O.: Information and Exponential Families In Statistical Theory. John Wiley and Sons (2014)
4. Beyer, H.G., Schwefel, H.P.: Evolution strategies - a comprehensive introduction. *Natural Computing* **1**(1), 3–52 (2002)
5. De Nobel, J., Vermetten, D., Wang, H., Shir, O.M., Emmerich, M., Bäck, T.: Integer natural evolution strategies - reproducibility and additional results (2026). <https://doi.org/10.5281/zenodo.19627585>
6. Hamano, R., Saito, S., Nomura, M., Shirakawa, S.: CMA-ES with margin: Lower-bounding marginal probability for mixed-integer black-box optimization. In: Proceedings of the Genetic and Evolutionary Computation Conference. pp. 639–647. GECCO '22, Association for Computing Machinery, New York, NY, USA (2022). <https://doi.org/10.1145/3512290.3528827>
7. Hamano, R., Saito, S., Nomura, M., Uchida, K., Shirakawa, S.: Catcma: Stochastic optimization for mixed-category problems. In: Proceedings of the Genetic and Evolutionary Computation Conference. p. 656–664. GECCO '24, Association for Computing Machinery, New York, NY, USA (2024). <https://doi.org/10.1145/3638529.3654198>
8. Hansen, N., Auger, A., Finck, S., Ros, R.: Real-parameter black-box optimization benchmarking: Experimental setup. Tech. Rep. (2012)
9. Hansen, N., Ostermeier, A.: Completely derandomized self-adaptation in evolution strategies. *Evol. Comput.* **9**(2), 159–195 (2001)
10. Ikeda, K., Ono, I.: Natural evolution strategy for mixed-integer black-box optimization. In: Proceedings of the Genetic and Evolutionary Computation Conference. pp. 831–838. GECCO '23, Association for Computing Machinery, New York, NY, USA (2023). <https://doi.org/10.1145/3583131.3590518>
11. Inusah, S., Kozubowski, T.J.: A discrete analogue of the laplace distribution. *Journal of Statistical Planning and Inference* **136**(3), 1090–1102 (2006)
12. Malagò, L., Matteucci, M., Pistone, G.: Towards the geometry of estimation of distribution algorithms based on the exponential family. In: FOGA 11. pp. 230–242. Association for Computing Machinery (2011)
13. Marty, T., Hansen, N., Auger, A., Semet, Y., Héron, S.: Lb+ic-cma-es: Two simple modifications of cma-es to handle mixed-integer problems. In: PPSN XVIII. pp. 284–299. Springer-Verlag (2024)
14. Murphy, K.: Machine Learning: A Probabilistic Perspective. Adaptive Computation and Machine Learning series, MIT Press (2012)
15. Ollivier, Y.: Online natural gradient as a kalman filter. *Electronic Journal of Statistics* **12**(2), 2930–2961 (2018)
16. Ros, R., Hansen, N.: A simple modification in cma-es achieving linear time and space complexity. In: International conference on parallel problem solving from nature. pp. 296–305. Springer (2008)
17. Rudolph, G.: An evolutionary algorithm for integer programming. In: Davidor, Y., Schwefel, H.P., Männer, R. (eds.) PPSN III. pp. 139–148. Springer Berlin Heidelberg (1994)

18. Rudolph, G.: Handbook of Natural Computing: Theory, Experiments, and Applications, chap. Evolutionary Strategies, pp. 673–698. Springer-Verlag (2012)
19. Shir, O.M., Emmerich, M.: Foundations of correlated mutations for integer programming. In: FOGA 18. pp. 285–296. FOGA’25, Association for Computing Machinery (2025)
20. Talbi, E.G.: Metaheuristics for variable-size mixed optimization problems: A unified taxonomy and survey. *Swarm and Evolutionary Computation* **89**, 101642 (2024)
21. Wierstra, D., Schaul, T., Glasmachers, T., Sun, Y., Peters, J., Schmidhuber, J.: Natural evolution strategies. *J. Mach. Learn. Res.* **15**(1), 949–980 (2014)